

**COMPARISON OF GPS TEC WITH IRI MODELS OF
2007, 2012, AND 2016 OVER SUKKUR, PAKISTAN.**



by

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APPROVAL BY BOARD OF EXAMINE

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DEDICATION

This research is dedicated to my Parents and my teachers who prayed for me and helped me to achieve my goals.

Adil Hussain

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All praises to Allah Almighty who is most beneficent and the most merciful.

Special thanks to my supervisor, Dr. Munawar Shah who helped me where I need.

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Adil Hussain

ABSTRACT

The international reference ionosphere (IRI) models has been widely used for specifying ionospheric parameters. An evaluation on the performance of the IRI models (version 2007, 2012 and 2016) over Sukkur, Pakistan (27.71° N, 68.85° E) is presented in this work. Total Electron Content (TEC) from IRI models and GPS TEC in the year 2019 over Sukkur region are compared in this work. Comparison of GPS with IRI models will help us to improve IRI modelling in Pakistan. This study will also help us to identify which IRI model is best correlated with GPS in this region for better use of IRI modelling in future. The development of more accurate TEC maps finds useful applications in enhancing the extent to which ionospheric influences on radio signals are corrected. GPS TEC and IRI model's data was collected on May 1, 2019 to May 3, 2019. Dst and Kp data was also compared in this work to estimate geomagnetic storm. Results of this study show very good correlations (above 0.83 for IRI 2016, above 0.82 for IRI 2012 and above 0.78 for IRI 2007) between the IRI-TEC predictions and the GPS-TEC measurements for the days examined, and so reveal the potential of the IRI model as a good candidate for an enhanced TEC modeling over Pakistan

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List of Abbreviations

GNSS	Global Navigation Satellite System
GPS	Global Navigation System
GALILEO	European Global Satellite Navigation System
GLONASS	Global Navigazionnaya Sputnikovaya Sistema
IRNSS	Indian regional navigation satellite system
DTEC	Difference in total electron content
USA	United states of America
EU	European union
TEC	Total Electron Content
SW	Solar Wind
SID	Sudden Ionospheric Disturbance
DST	Disturbance storm time

1 Introduction

Navigation improvement is always the key issue for mankind for betterment in everyday life and for research purpose. Man is working on navigation improvements for decades. In this chapter a detail analysis of different integrities of GNSS their applications are summarized in all aspects. Furthermore, it will provide deep insights in the GNSS technology for its use in different technologies. [1]

1.1 Introduction to GNSS

What has most significantly changed navigation techniques is the advent of Global Navigation Satellite Systems (GNSS), which started with the launch of the U.S. Department of Defense Global Positioning System (GPS) in the late 1970s. GPS was launched in late 70s after different experiments in this field. After that a revolution is seen in the field of navigation. [2]

Global Navigation Satellite System (GNSS) refers to a constellation of satellites providing signals from space that transmit positioning and timing data to GNSS receivers. The receivers then use this data to determine location. By definition, GNSS provides global coverage. Early applications of GNSS were developed for the military and soon expanded to the survey and mapping industries—driven largely by the tremendous advances in accuracy, efficiency as well as cost reductions. [3]

Now, vehicles, whether on land, in the air or at sea—routinely rely on the precise positioning information provided by GNSS technology. In fact, the ready adoption of the technology, from

mining to unmanned, and the increasingly complex requirements for positioning, anywhere, anytime, is driving innovation in the industry that includes the integration of GNSS technology with a variety of other sensors and methodologies. This multi combination approach is sure to drive innovation in the industry for many years to come. [4]

GNSS Systems

Although you may already be familiar with the term “GPS” (Global Positioning System), here is another system called “GNSS” which is used to describe the collection of satellite positioning systems that are now operating or planned. The main satellite missions in GNSS.

GPS (United States of America): GPS was the first mission in the GNSS system. GPS was launched in the late 1970s by the United States department of Defense. It uses a constellation of 27 satellites and provides global coverage. [5]

GLONASS (Russia): GLONASS is operated by the Russian government. The GLONASS constellation consists of 24 satellites and provides global coverage.

Galileo (European Union): Galileo is a civil GNSS system operated by the European Global Navigation Satellite Systems Agency (GSA). Galileo will use 27 satellites with the first Full Operational Capability (FOC) satellites being launched in 2014.

BeiDou (China): BeiDou is the Chinese navigation satellite system. The system will consist of 35 satellites. A regional service became operational in December of 2012. [6]

IRNSS (India): The Indian Regional Navigation Satellite System (IRNSS) provides service to India and the surrounding area.

QZSS (Japan): QZSS is a regional navigation satellite system that provides service to Japan and the Asia-Oceania region. [7]

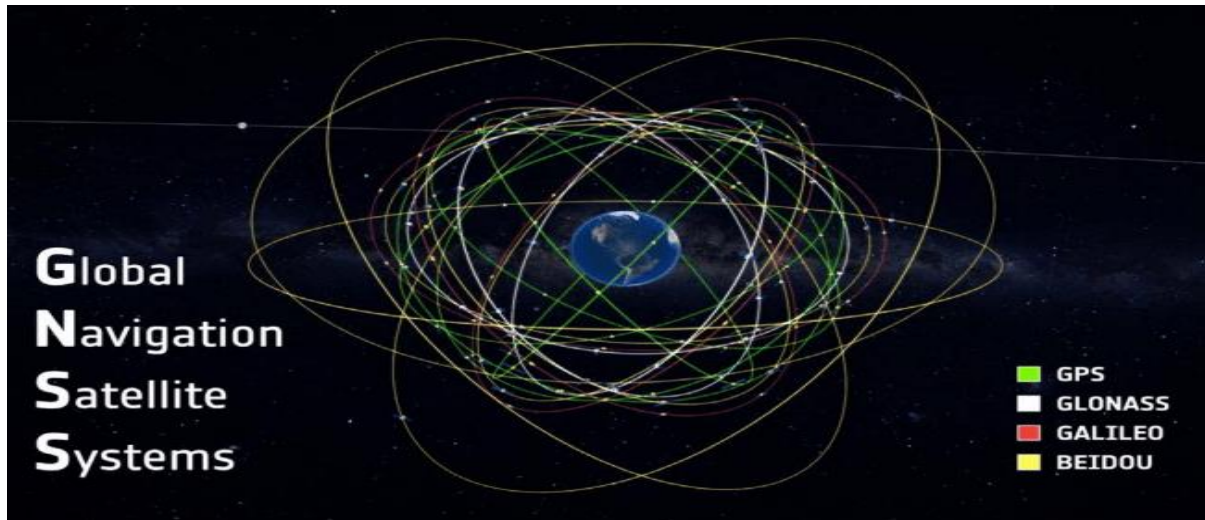


Figure 1.1: a view of different GNSS systems

1.2 Total Electron Content (TEC)

The Total Electron Content (TEC) is the total number of electrons present along a path between radio transmitter and receiver. Radio waves are affected by the presence of these electrons. The more electrons in the path of the radio wave, the more the radio signal will be affected. For ground to satellite communication and satellite navigation, TEC is a good parameter to monitor for possible space weather impacts. TEC is measured in electrons per square meter. By

convention, 1 TEC Unit TECU = 10^{16} electrons/m². Vertical TEC values in Earth's ionosphere can range from a few to several hundred TECU. [8]

The TEC in the ionosphere is modified by changing solar Extreme Ultra-Violet radiation, geomagnetic storms, and the atmospheric waves that propagate up from the lower atmosphere. The TEC will therefore depend on local time, latitude, longitude, season, geomagnetic conditions, solar cycle and activity, and troposphere conditions. The propagation of radio waves is affected by the ionosphere. [9] The velocity of radio waves changes when the signal passes through the electrons in the ionosphere. The total delay suffered by a radio wave propagating through the ionosphere depends both on the frequency of the radio wave and the TEC between the transmitter and the receiver. At some frequencies the radio waves pass through the ionosphere. At other frequencies, the waves are reflected by the ionosphere. [10]

The change in the path and velocity of radio waves in the ionosphere has a big impact on the accuracy of satellite navigation systems such as GPS/GNSS. Neglecting changes in the ionosphere TEC can introduce tens of meters of error in the position calculations. The Global Positioning System (GPS), the US part of GNSS, uses an empirical model of the ionosphere, the Klobuchar model, to calculate and remove part of the positioning error caused by the ionosphere when single frequency GPS receivers are used. [11]

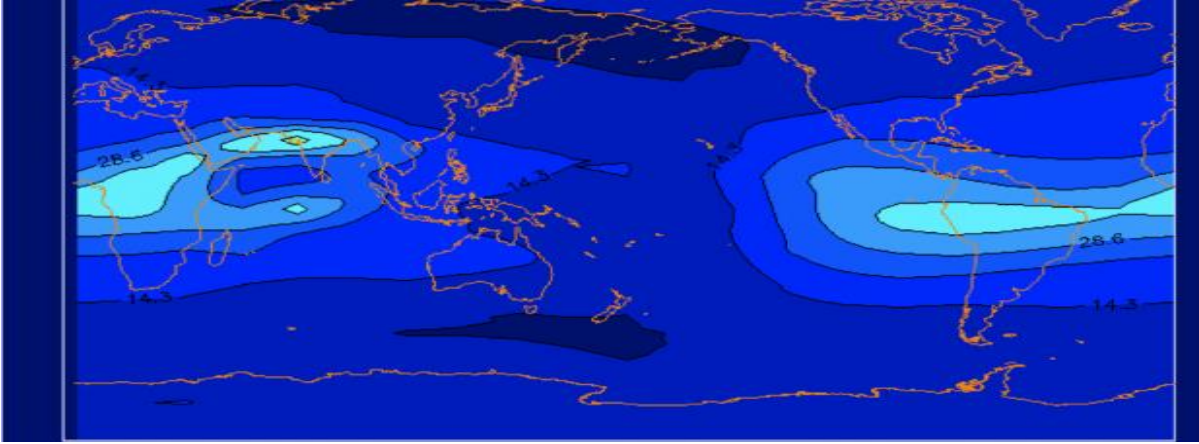


Figure1.2: Spatial TEC map from GNSS over the world for every two hours. The temporal resolution is bi hour, while spatial extension is 5.0 x 2.5 in longitude and latitude (Courtesy IGS GNSS)

1.3 Space Weather

Activity on the Sun's surface creates a type of weather called space weather. The Sun is far away—about 93 million miles (150 million kilometers) from Earth. However, space weather can affect Earth and the rest of the solar system. At its worst, it can even damage satellites and cause electrical blackouts on Earth. [12]

Scientists are studying space weather using different tools nowadays. One of these tools is GNSS including GPS, which we already discussed in above section.

GNSS DELAYS

Errors in GNSS are due to: [13]

- **Solar activity** like geomagnetic storm as a result of significant changes in the speed of solar wind(gust). Magnetospheric sub storm, sudden ionospheric disturbance (SID), ionospheric storm, travelling ionospheric disturbance (TID), scintillations.

- Ionosphere delay ionosphere is one of the major sources of GNSS signal delays.
- Troposphere delay
- Multipath

1.4 GEOMAGNETIC STORMS

A geomagnetic storm is a temporary disturbance of the Earth's magnetosphere caused by a solar wind shock wave and/or cloud of magnetic field that interacts with the Earth's magnetic field.

The disturbance that drives the magnetic storm may be a solar coronal mass ejection or a co-rotating interaction region (CIR), a high-speed stream of solar wind originating from a coronal hole. The frequency of geomagnetic storms increases and decreases with the sunspot cycle. During solar maximum, geomagnetic storms occur more often, with the majority driven by CMEs. During solar minimum, storms are mainly driven by CIRs. [14]

The increase in the solar wind pressure initially compresses the magnetosphere. The solar wind's magnetic field interacts with the Earth's magnetic field and transfers an increased energy into the magnetosphere. Both interactions cause an increase in plasma movement through the and an increase in electric current in the magnetosphere and ionosphere. During the main phase of a geomagnetic storm, electric current in the magnetosphere creates a magnetic force that pushes out the boundary between the magnetosphere and the solar wind. [15]

Several space weather phenomena tend to be associated with or are caused by a geomagnetic storm. These include solar energetic particle (SEP) events, geomagnetically induced currents (GIC), ionospheric disturbances that cause radio and radar scintillation, disruption of navigation by magnetic compass and auroral displays at much lower latitudes than normal. [16]

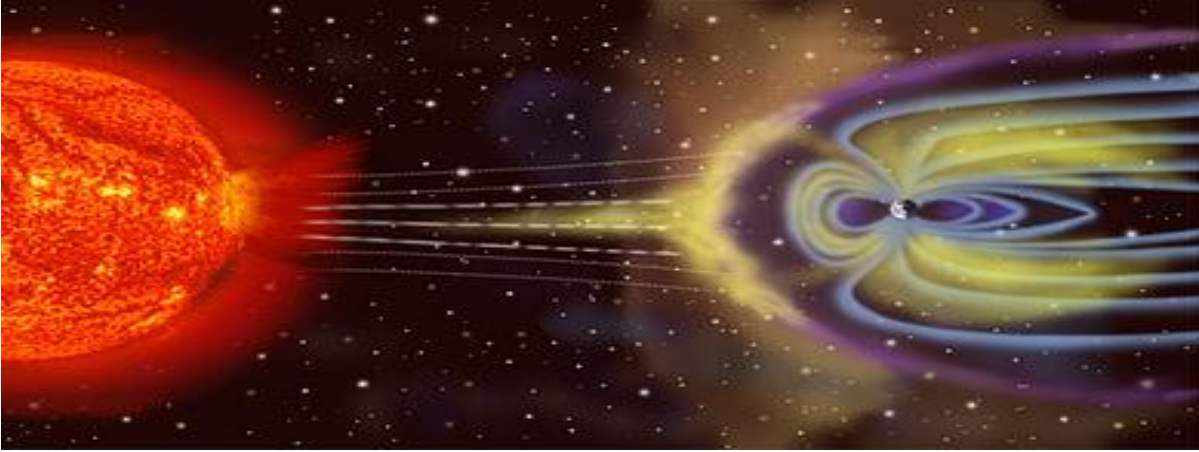


Figure: Geomagnetic storm and effects on earth's magnetic field (Courtesy NASA)

1.5 IONOSPHERIC DELAY

The ionosphere is a part of the earth's atmosphere, which extends from approximately 50 – 1000 km in altitude. The sun's ultraviolet and x-ray emissions are primarily responsible for causing ionization in the ionosphere. When a signal from the GNSS satellite is travelling through the ionized layer in the ionosphere, refraction, or bending of the wave, occurs. The amount of refraction that occurs depends on three main factors: (1) the density of ionization of the layer, (2) the frequency of the radio wave, e.g. for GPS L1 it is 1575.42 MHz, and (3) the angle at which the wave enters the layer.

An important descriptive quantity in describing the effect of the ionosphere on the GNSS signal is the total electron content (or TEC). TEC is the total number of electron present along a path between the satellite and the receiver on earth, with units of electrons per square meter, where 10^{16} electrons/m² = 1 TEC unit (TECU). [17]

1.6 Tropospheric Delay

Troposphere is the atmospheric layer placed between earth's surface and an altitude of about 60km. The effect of troposphere on the GNSS signals appears as an extra delay in the

measurement of the signal traveling from the satellite to receiver. This delay depends on the temperature, pressure, humidity and transmitter and receiver antennas location. The main feature of the troposphere is that it is a nondispersive media with respect to electromagnetic waves up to 15GHz. [18] The tropospheric effects are not frequency dependent for the GNSS signals; hence the carrier phase and code measurements are affected by the same delay. An immediate consequence of being a non-frequency dependent delay is that the tropospheric refraction can not be removed by combinations of dual frequency measurements. The only way to reduce tropospheric effect is to use models and to estimate it from observational data. Nevertheless, fortunately, most of the tropospheric refraction comes from the predictable hydrostatic component. [19]

A brief description on the atmosphere dry and wet component effects on the GNSS signals is given as follows:

Dry component delay: It is caused by the dry gases present at the troposphere (78% N₂, 21% O₂ and 0.9% Ar...). Its effect varies with local temperature and atmospheric pressure in quite a predictable manner, as its variation is less than that the 1% in a few hours. The error caused by this component is about 2.4 meters in the zenith direction and 10 meters for lower elevations. [20]

Wet component delay: It is caused by the water vapour and condensed water in form of clouds. It depends on weather conditions. The excess delay is small in this case, only few tens of centimetres, but this component varies faster than the hydrostatic component and a quite randomly way, being very difficult to model. The dry atmosphere can be modelled from surface

pressure and temperature using the laws of the ideal gases. The wet component is more unpredictable and difficult to model, thence for high precision navigation, this delay is estimated together with the coordinates. [21]

The tropospheric delay depends on the signal path through the neutral atmosphere, and thence, can be modelled as a function of the satellite elevation angle. Due to the differences between the atmospheric profiles of the dry gases and water vapour it is better to use different mappings for the dry and wet components. [22]

Several nominal tropospheric models are available in the literature, which differ on the assumptions made on the vertical profiles and mappings. Basically, they can be classified in two main groups: Geodetic-oriented or Navigation-oriented. The first group Sastamoinen, Hopfield, among others are more accurate but generally more complex, and need surface meteorological data, being their accuracy affected by the quality of these data. The second group are less accurate, but meteorological data are not needed. [23]

1.7 MULTIPATH DELAY IN GNSS SIGNALS

In radio communication, multipath is a phenomenon that results in radio signals reaching the receiver antenna by two or more paths. Causes of multipath include ionospheric reflection and refraction, and reflection from water bodies and terrestrial objects such as mountains and buildings. [24]

Multipath propagation causes multipath interference, including constructive and destructive interference and phase shifting of the signal. Ddestructive interference

causes fading. This may cause radio signal to become weaker in certain areas to be received a. So multipath propagation can be detrimental in radio communication systems. Where the magnitudes of the signals arriving by the various paths have a distribution known as the Rayleigh distribution, this is known as Rayleigh fading. [25]

2 DATA AND METHODOLOGY

2.1 Introduction to ionosphere.

The ionosphere is the region of our atmosphere characterized by the presence of ionized plasma. The region extends in altitudes of about 50 km to 1000 km from the Earth's surface and is characterized by the presence of an amount of ionized plasma that can have substantial influence on radio propagation through it. It is a region that includes the thermosphere and parts of the mesosphere and exosphere. The ionosphere is ionized by solar radiation. [26] It plays an important role in atmospheric electricity and forms the inner edge of the magnetosphere. It has practical importance because, among other functions, it influences radio propagation to distant places on the Earth. it exist primarily due to ultraviolet radiation from the sun. Following figures are showing relationship of ionosphere with atmospheric layers and ionospheric graphic.[27]

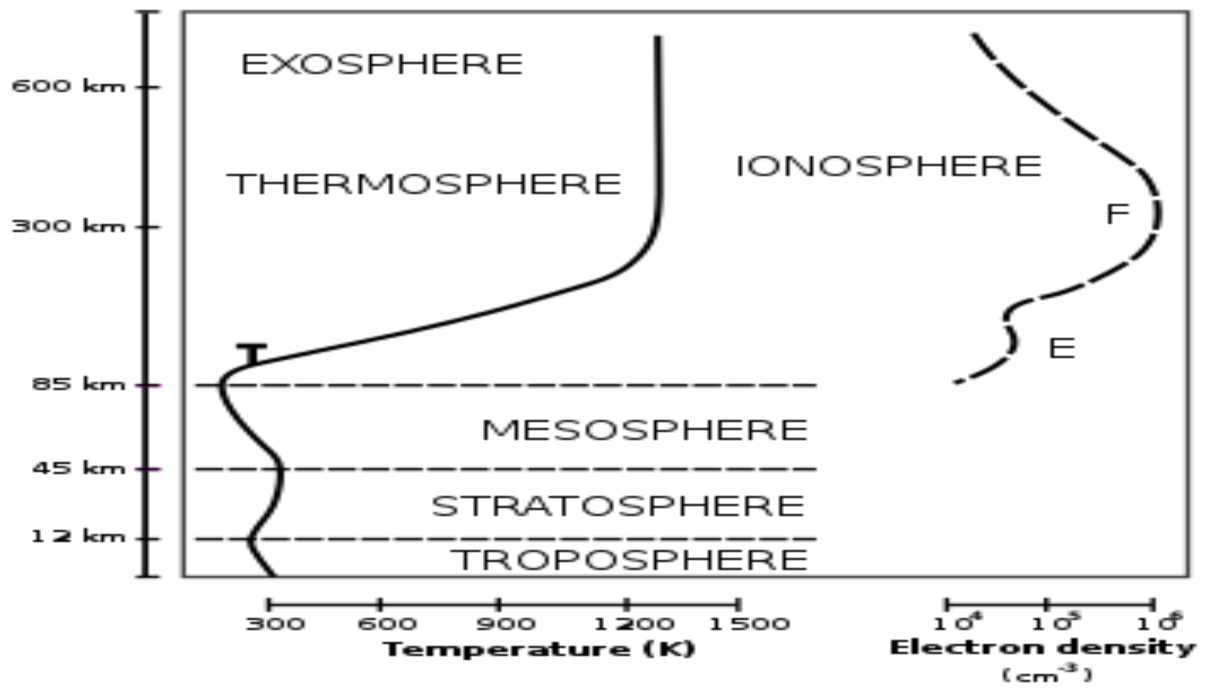


Figure 2.1: Relationship of the atmosphere and ionosphere

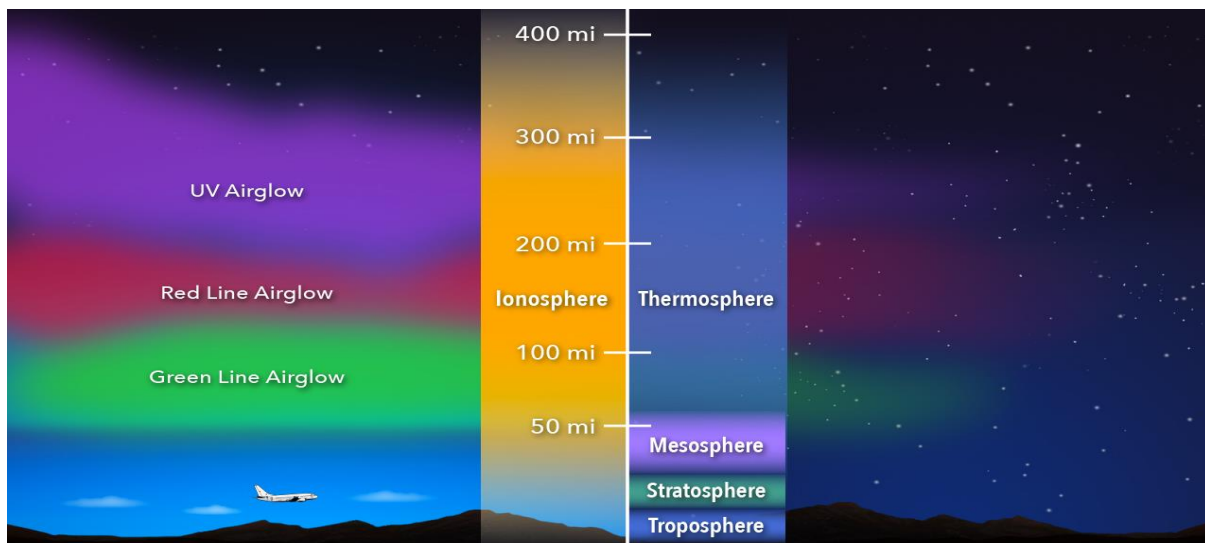


Figure 2.2: Ionosphere graphics

2.2 Study area and methodology

Due to the ionosphere's dispersive nature, radio signals propagating through it experience frequency-dependent group delays and phase advances [Opperman et al., 2007]. Increased knowledge of the ionosphere, especially for lesser known areas like the African equatorial regions, will help us correct for influences that the ionosphere has on radio signals propagating through it. The International Reference Ionosphere (IRI) model has been accepted as a defector standard for specifying ionospheric parameters across the globe. In this thesis, we present an evaluation of the model's performance over Sukkur, Pakistan. Sukkur is a low latitude region and Pakistan's very first multifrequency GPS receiver is installed in IBA university, ring chowk Sukkur. [28]



Figure 2.3: Study area map showing Geographical location of Sukkur, Pakistan.

The geographic and geomagnetic coordinates of Sukkur are 27.71° N, 68.85° E. The town lies within the low to moderate Equatorial region. The Global Positioning System (GPS) is a satellite-based navigation system consisting mainly of a network of transmitter satellites and receivers. Each of the transmitter satellites continually beep out radio signals from space containing their respective 3-D positions and the time of transmission. A receiver on Earth will usually need to receive beeps from at least 4 of the transmitter satellites to be able to use in-built computer programs to compute its (the receiver's) location and current time. [29] Given the GPS satellite orbital design considerations, a total of 24 operational transmitter satellites are

required to form an evenly distributed constellation round about the globe to ensure that at least 4 of them are visible over any given location on earth. Dual-frequency receivers are designed to be able to receive signals from the transmitter satellites in two bands (L1 band - 1.57542 GHz, and L2 band – 1.2276 GHz). [30] This makes it possible to include algorithms within such receivers to compute ionospheric delays caused to the propagating radio signals in the form of TEC since the GPS signals transverse the ionosphere carrying signatures of the dynamic medium and thus offer opportunities for ionospheric research [Bhuyan and Rashmi, 2007]. In this work, we compare GPS TEC with the IRI models with TEC observations of GPS receiver at Sukkur, Pakistan. We also look forward that this work and future related work on this region will provide useful information on a proposed TEC modeling over Pakistan using the IRI models. [31]

2.3 Vertical TEC from the IRI model

The International Reference Ionosphere (IRI) is a working group, jointly sponsored by the Committee on Space Research (COSPAR) and the International Union of Radio Science (URSI), that develops an empirical ionospheric model called the IRI model. The IRI model gives spatial and temporal representations of ionospheric parameters (including TEC) and has been widely regarded as the international standard for specifying ionospheric parameters [Bilitza, 2001]. [32] Bilitza and Reinisch [2008] also noted that the IRI model has become so widely accepted. In this work, we use the IRI-2007, IRI-2012 and IRI 2016 model's data that is taken in 1st, 2nd and 3rd of May 2019. In general, the standard IRI options were used, and the upper integration limit for the TEC values was set at 1000km, a good limit for the IRI models [Bilitza corresponding data from the GPS equipment also excludes plasmaspheric TEC

contribution. and McKinnell, 2011], and to clearly exclude TEC contribution from the plasmasphere, since [33]

2.4 Vertical TEC from GPS Receivers

Figure 2.3 depicts the relationship between slant and vertical TEC. To compare the electron contents for paths with different elevation angles, the TECs must be transformed into equivalent vertical content or vertical TEC by dividing it by the secant of the elevation angle at a mean ionospheric height. [34]

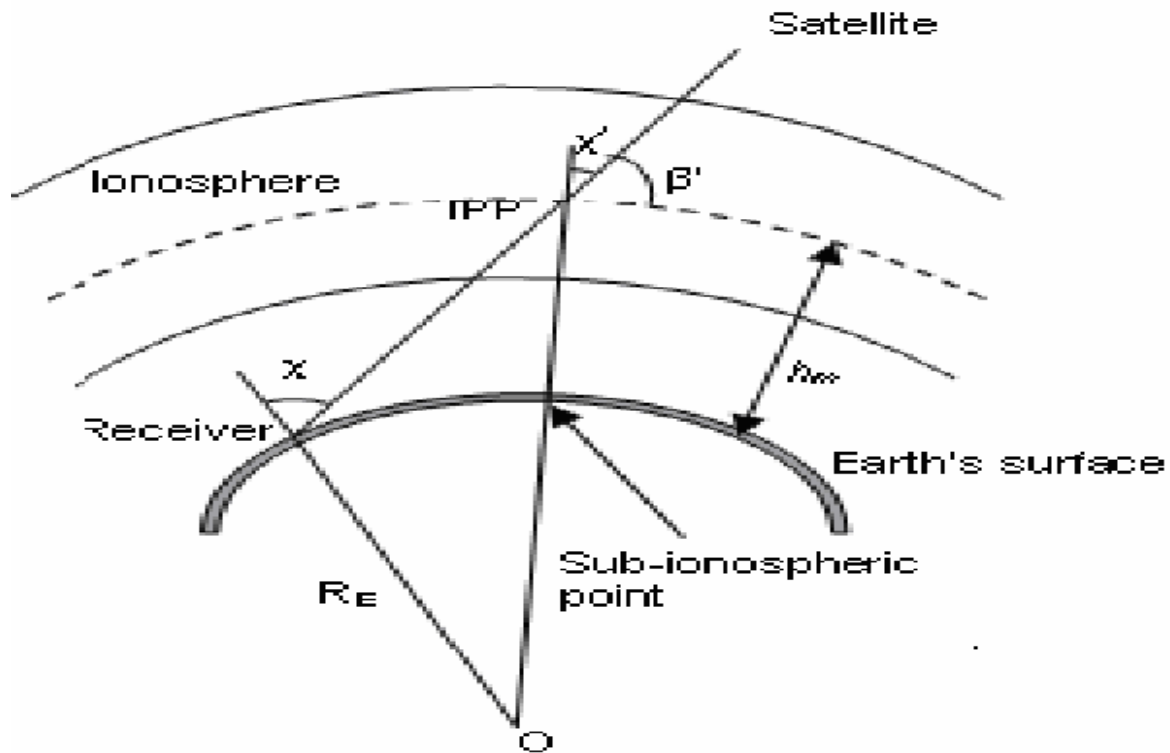


Figure 2.3: single layer model TEC

This study utilized the GPS-TEC data derived from dual frequency GNSS receiver operating at Sukkur Station in Pakistan during years 2019. This GNSS station is located at Sukkur (27.71°N, 68.85° E). The GPS data are logged in Receiver Independent Exchange (RINEX) format, where Slant TEC (STEC) can be obtained along the line of sight between satellite and ground station. It is calculated as the total number of free electrons in a (1×1) square meter column. It is measured in TECU and can be computed from the following equations (Jin et al. 2014; Tariq et al. 2019b) [35]

$$\text{STEC} = \frac{f_1^2 f_2^2}{40.28 (f_1^2 - f_2^2)} (L_1 - L_2 + \lambda_1 (N_1 + b_1) - \lambda_2 (N_2 + b_2) + \epsilon) \quad (1)$$

$$\text{STEC} = \frac{f_1^2 f_2^2}{40.28 (f_1^2 - f_2^2)} (p_1 - p_2 - (d_1 - d_2) + \epsilon) \quad (2)$$

where STEC is the Slant Total Electron Content, f1 and f2 are carrier-phase frequencies of GPS signals at two ends, respectively. L represents the carrier-phase observations of the delay signals as denoted by L1 and L2, while P represents the pseudo-range measurements of the delay path denoted by P1 and P2. λ is the wavelength of GPS signals. The ambiguity of the signal is N, while b and d are the instrumental and differential code biases. Moreover, STEC can be converted into Vertical TEC (VTEC) by using the equation (Klobuchar 1987). [36]

$$\text{VTEC} = \text{STEC} \times \cos \left(\arcsin \left(\frac{R \sin Z}{R + H} \right) \right) \quad (3)$$

where R represents the Earth's radius, H is the height of ionospheric upper layer, while Z is the satellite elevation angle from the point of observation (Yuan et al. 2008; Li et al. 2015; Barkat et al. 2018). However, VTEC values from the GPS observational RINEX files from station are processed independently by STEC calibration method (Ciraolo et al. 2007). The data files have been provided by Rasim Shehzad (IST). The satellite elevation angle is set at about $> 20^\circ$ to reduce the multipath effects and then VTEC values are retrieved in hourly temporal resolution. [37]

$$\text{correlation coefficient} = \frac{\sum (GPSTEC_i - \overline{GPSTEC_i}) (IRITEC_i - \overline{IRITEC_i})}{\sqrt{\sum (GPSTEC_i - \overline{GPSTEC_i})^2} \sqrt{\sum (IRITEC_i - \overline{IRITEC_i})^2}} \quad (4)$$

Correlation coefficient is measured using equation 4 for GPS TEC all IRI models and then data is compared. In order to evaluate the model performance in terms of daily fluctuations, the diurnal hourly VTEC are selected for three days in the month of May at the Sukkur stations from in 2019. Also, to analyze the daily VTEC fluctuations, the measured and modelled hourly VTEC for every day are added and averaged. [38]

DTEC and percentage deviation were also calculated to find mean difference and percentage. Following formulas are used to calculate DTEC and percentage deviation. [39]

$$d \text{ TEC} = \text{IRI TEC} - \text{GPS TEC} \quad \text{-----} \quad 5$$

$$pd = \frac{(IRITEC_i) - (GPSTEC_i)}{GPSTEC_i} \times 100 \quad \text{-----} \quad 6$$

3 RESULTS AND DISCUSSIONS

Comparison was made between GPS and all IRI models over Sukkur region to identify the advantages of these models and for better use of these models in future over this region. Methods to get required data for this work are discussed and finalized in chapter 2. After complete analysis following are the results of this work.

3.1 Full comparison of GPS with IRI models

Hour to hour variation in Figure 3.1 shows the relation between IRI models TEC and GPS TEC over Sukkur region in first three days of May 2019. Maximum TEC values were observed between the time of sunrise and sunset (about 08:00 hours to 07:00 hours) of local time in all three days.

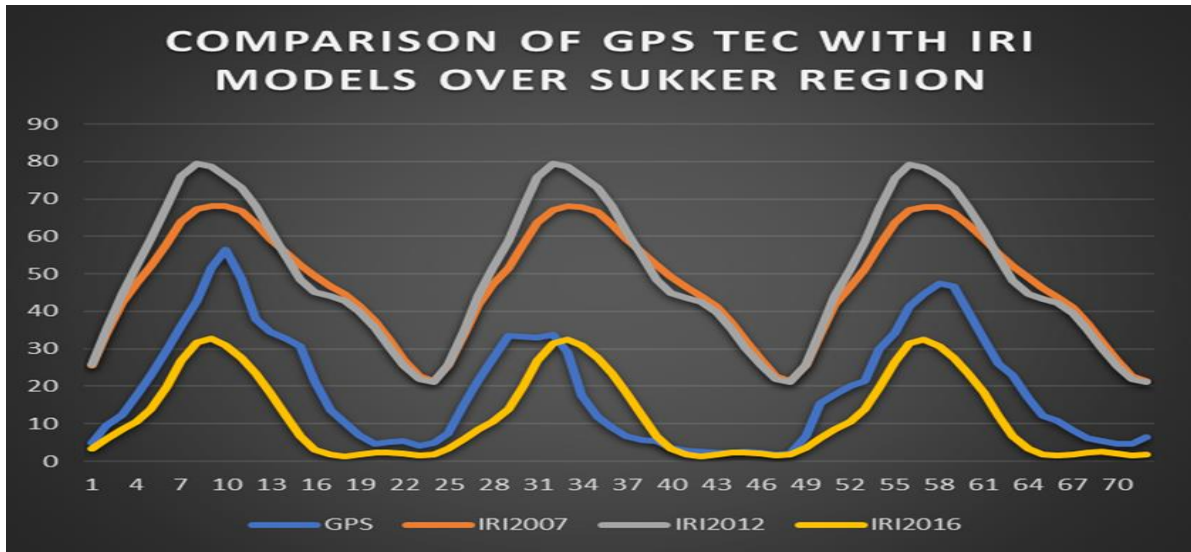


FIGURE 3.1: COMPARISON OF GPS WITH IRI MODELS MAY 1,2 AND 3, 2019

Minimum values were observed (between 08:00 hours and 01:00 hours) of local time. It is affirmation that the solar radiation is a major cause of the ionization process in the ionosphere. The observations generally agree with well-known ionospheric theories on ionization [40]

3.2 Correlation of GPS with IRI models

Correlation between GPS TEC and IRI models is shown in table 3.1. GPS TEC is best correlated with IRI 2016 model (above 0.83) and least correlated with IRI 2007 (above 0.78).

Table 3.1. Correlation values of GPS TEC with IRI models

Sr. No	Correlation of GPS	Correlation value
A	WITH IRI 2007	0.787797
B	WITH IRI 2012	0.827017
C	WITH IRI 2016	0.830174

Figure 3.2 is plot between different correlation values. Correlation of GPS with IRI 2012 and IRI 2016 are almost overlapping each other but correlation of GPS with IRI 2007 is little bit below in the graph showing that correlation of GPS with IRI 2012 and IRI 2016 are having good correlation coefficient. remember that table 3.1 is showing only averaged value and figure 3.2 is showing the plot of hourly data of first three days of May 2019. [41]

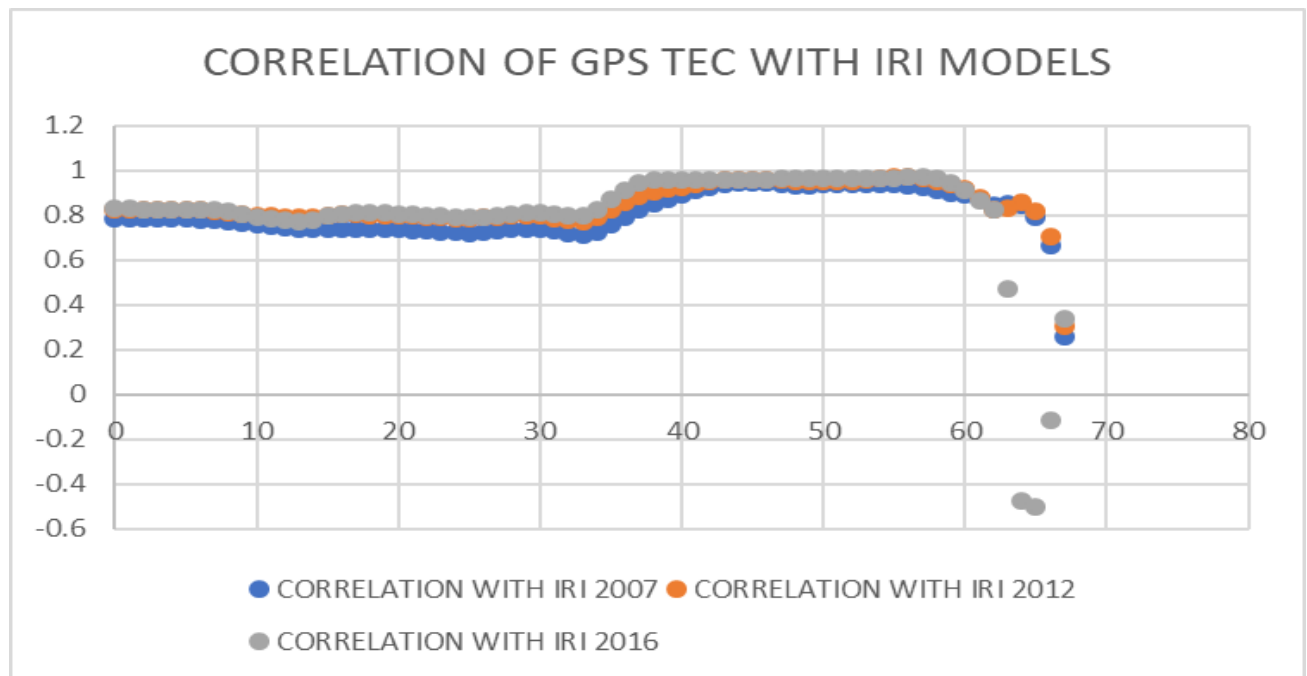


Figure 3.2: correlation graph of GPS TEC with IRI models

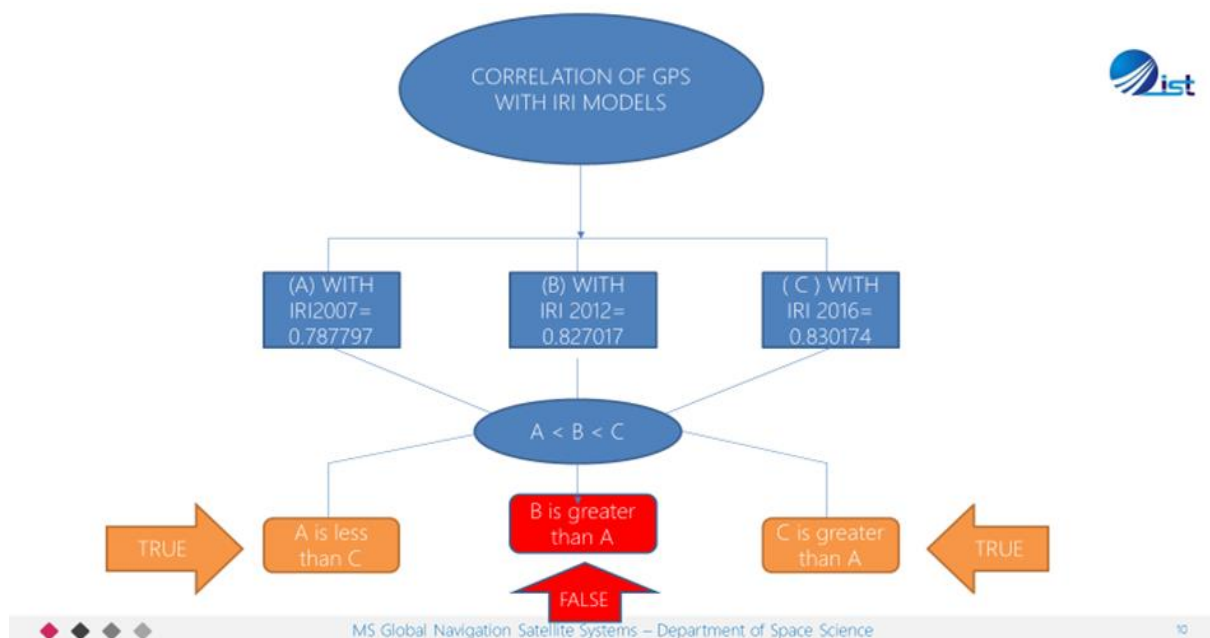


Figure 3.3. Flow chart showing correlation of GPS with IRI models

Table 3.1 shows the correlation values of GPS TEC with three IRI models. Data shows very good correlation of GPS with IRI models specially IRI 2016 (above 0.83). correlation of GPS with IRI 2012 is above 0.82 and correlation of GPS with IRI 2007 is above 0.78. figure 3.3 also showing the comparison of correlation values. Light discrepancies are due to upper integration limit which was taken 1000km. As we know that upper integration limit is not constant, and the limits will not always correspond to 1000km as we have used in this work. Here is another possible reason for the discrepancies is in the relatively small data from the region considered in developing the model. Given that the IRI model is databased, the accuracy of the model in a specific region depends on the availability of reliable data for the region. [42]

3.3DTEC

Difference of IRI TEC and GPS TEC is shown in figure 3.4. As per this graph IRI 2007 – GPS TEC is showing maximum values. It means that correlation of GPS TEC with IRI 2007 is less as compared to other models. From figure we can also see that IRI 2016 is at lowest difference with GPS. Averaged values of DTEC are shown in table 3.2. [43]

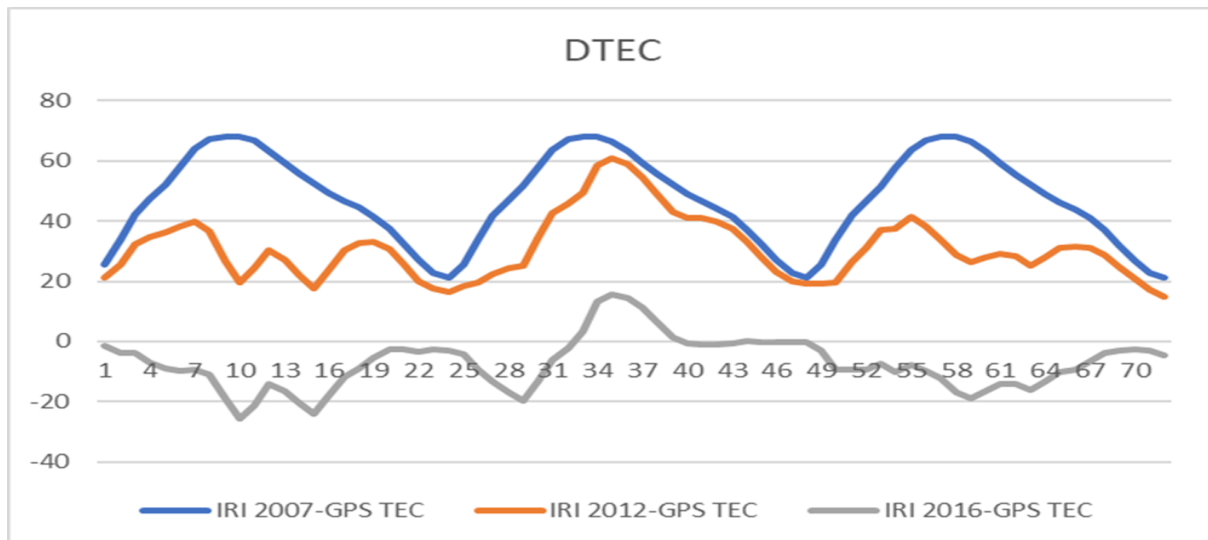


Figure 3.4: DTEC (IRI – GPS)

3.4 Percentage Deviation

Percentage deviation graph (figure 3.5) is an affirmation that IRI 2016 is at least percentage deviation. Maximum percentage deviation goes to IRI 2007 and IRI 2012 is also having greater percentage deviation as compared to IRI 2016. Averaged values percentage deviation are also shown in table 3.2. [44]

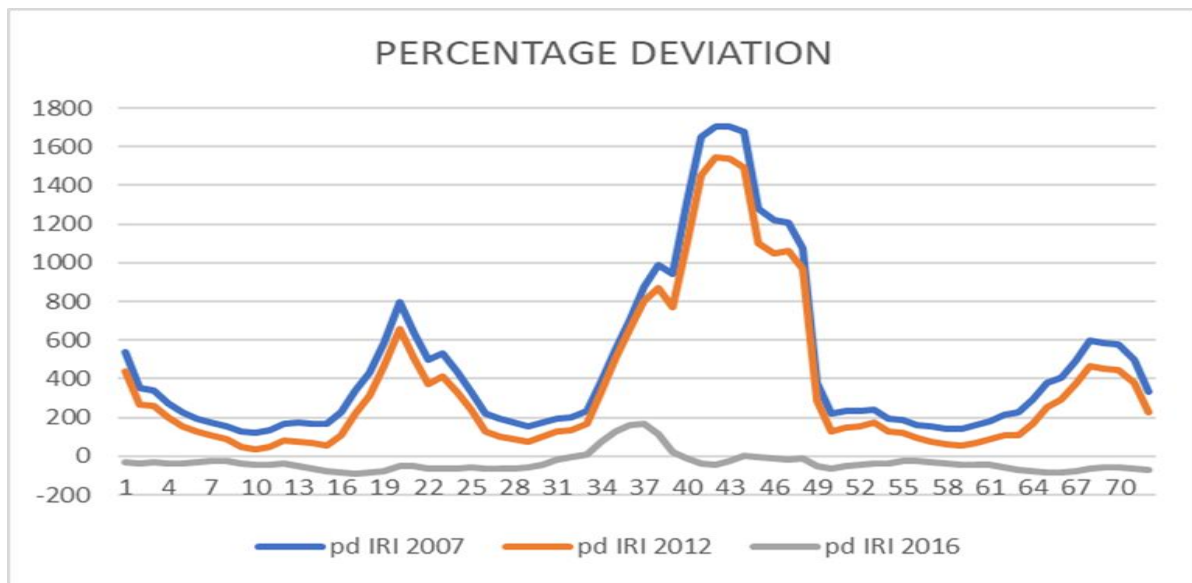


Figure 3.5: percentage deviation

Table 3.2: DTEC and percentage deviation

DTEC and Percentage Deviation



Sr. No	IRI models	DTEC with GPS (Averaged)	percentage deviation (Averaged)
1	IRI 2007	47.64305556	475.3594
2	IRI 2012	30.98111	371.8722
3	IRI 2016	-7.07861	-32.6402

3.5 Mean Values and Standard Deviation

TABLE 3.3: STATISTICAL ANALYSIS OF GPS AND ALL IRI MODELS

STATISTICAL ANALYSIS



SOURCE	MEAN VALUE	N	STD. DEVIATION	MINIMUM	MAXIMUM
GPS	19.18	72	4.864945897	1.8754896	56.483883
IRI 2007	47.64306	72	17.18875601	21.4	68.1
IRI 2012	50.1625	72	17.62989885	21.2	79.5
IRI 2016	12.10278	72	3.011644069	1.4	32.8



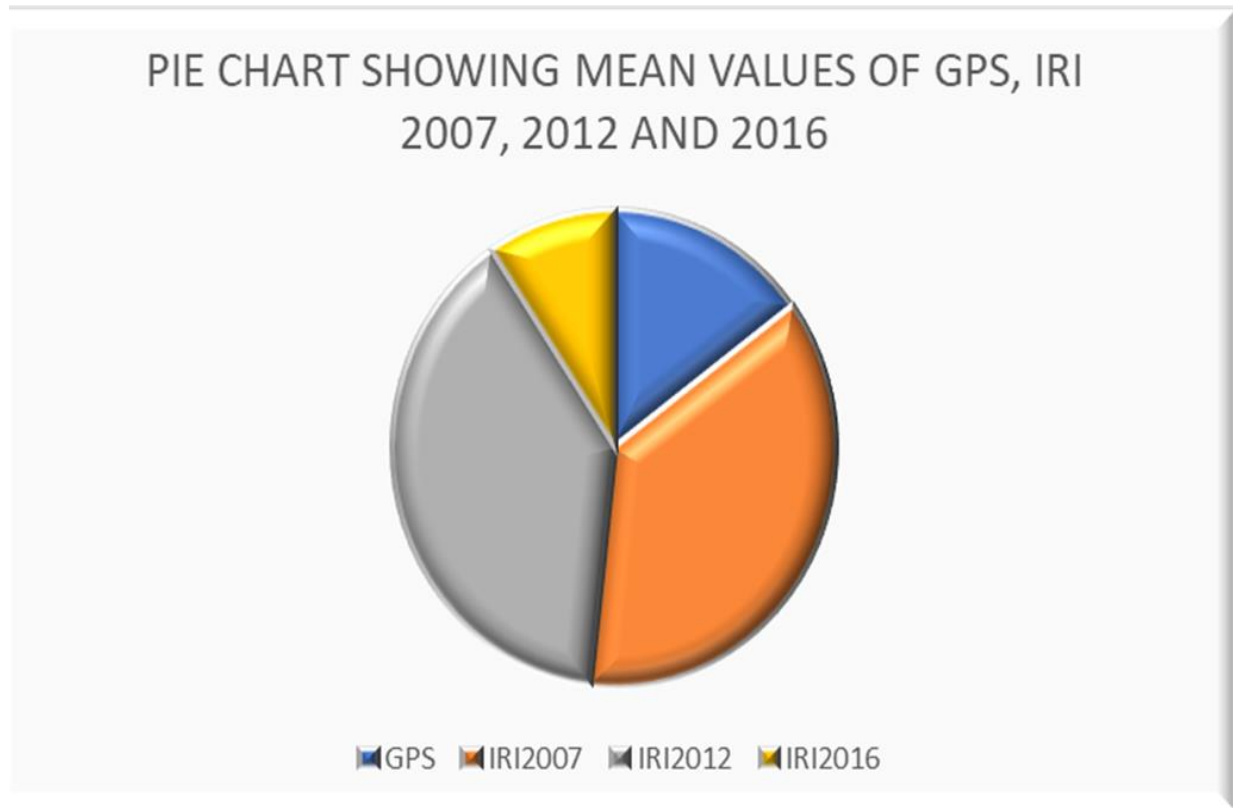


FIGURE3 .6: MEAN VALUES OF GPS AND IRI MODELS

Figure 3.6 and table 3.3 shows the mean of three days hourly data of GPS, IRI 2007, IRI 2012 and IRI 2016. According to this information IRI 2012 has the maximum mean value (50.1625) as compared to others. Moreover, IRI 2016 model has minimum value 12.10278 which is closest to that of GPS mean value which stands at 19.18. Mean TEC value of IRI 2007 is 47.64306. Standard deviation is also calculated and shown in table 3.2. GPS and IRI 2016 are closest in standard deviation and IRI 2007 and IRI 2012 are closest to each other.

3.6 Plasma Speed and Storm Indices

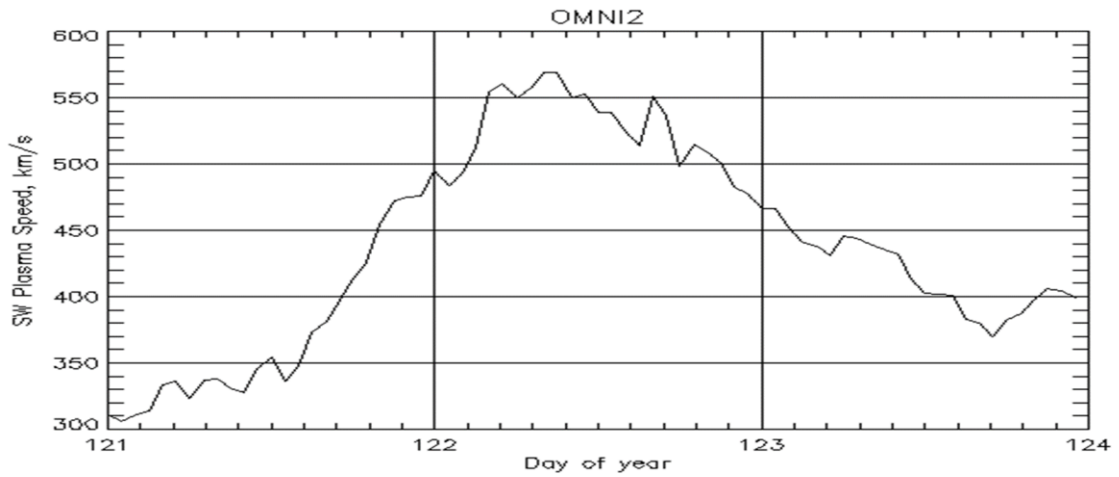


Figure 3.7: Plasma Speed In 121, 122 And 123 Days Of 2019

SW Plasma speed graph is shown in figure 3.7. Maximum value of plasma speed was on 2nd May 2019. 1st of May 2019 is showing least values. Maximum value of plasma speed was above 550 km / s. and minimum value was about 300 km / s. [45]

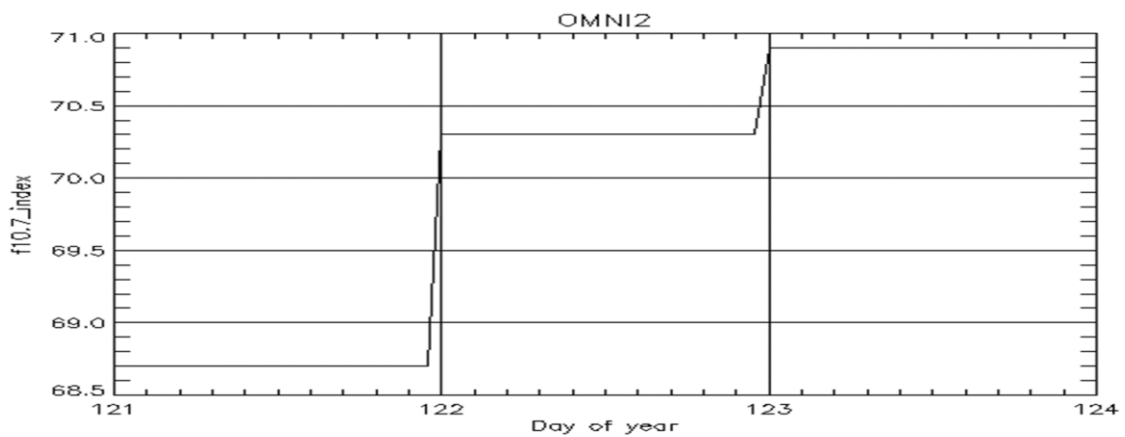


Figure 3.8: F 10.7 INDEX PLOT

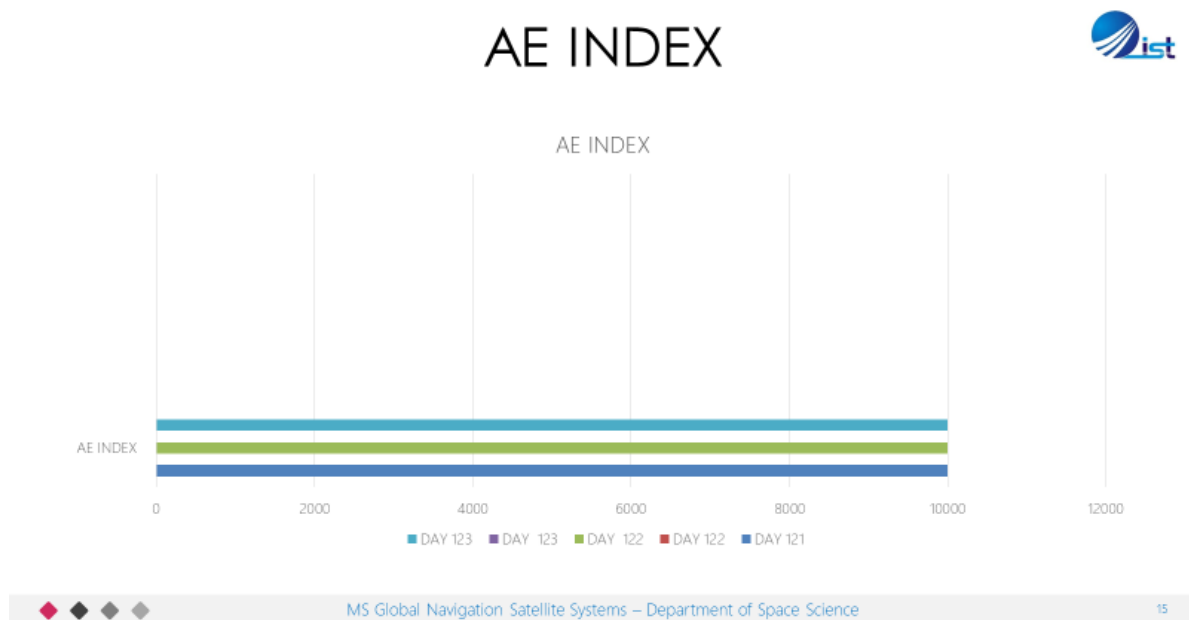


Figure 3.9: ae index value of day 121, day122 and day 123 ,2019

Figures 3.8 and 3.9 are showing solar index f10.7 and AE index, NT respectively.

maximum value of f10.7 was recorded on 123rd day of the year 2019 and minimum value was recorded on 121st of the year 2019. AE index was showing same value 9999 in all three days.

[46]

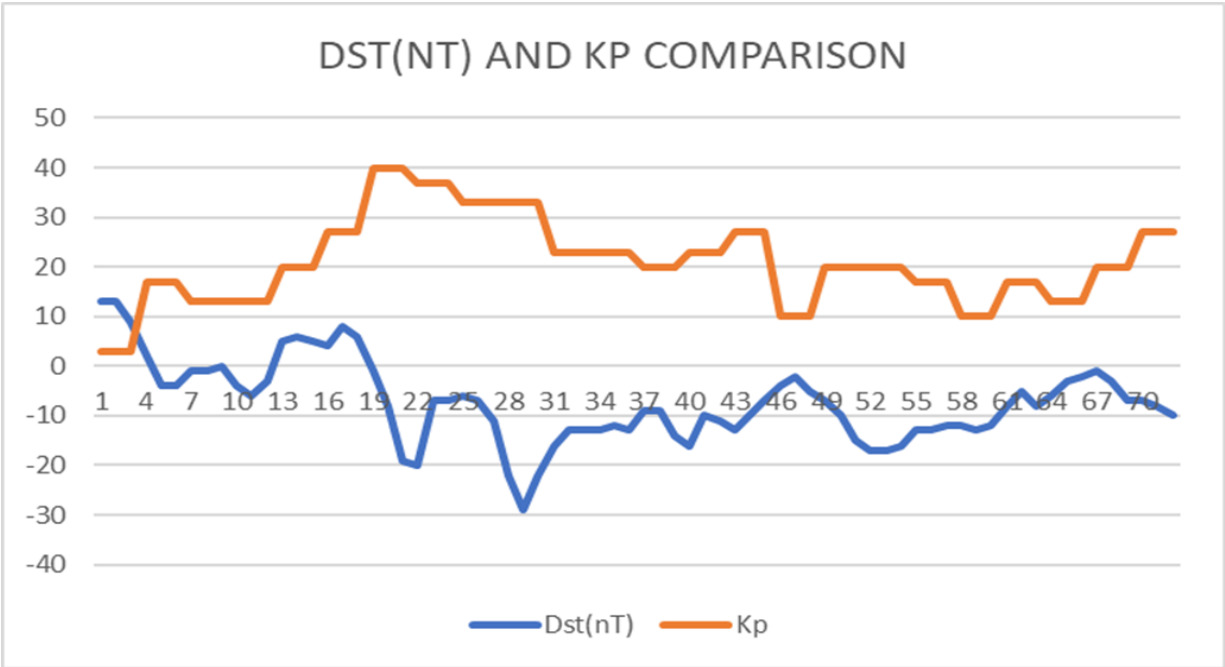


Figure3.6: Dst and Kp Indices Comparison.

Comparison between Dst and Kp indices have been made in figure 3.6. Dst and Kp index estimate the strength of storm. Both values in above mentioned graph are showing the greater strength of storm from 7pm May 1, 2019 to 5am May 2, 2019. Moderate strength is shown at the daytime on May 2, 2019. An increase in strength of storm is showing in the closing hours of 3rd May 2019. [47]

3.7 O/N₂ (COLUMN DENSITY RATIO)

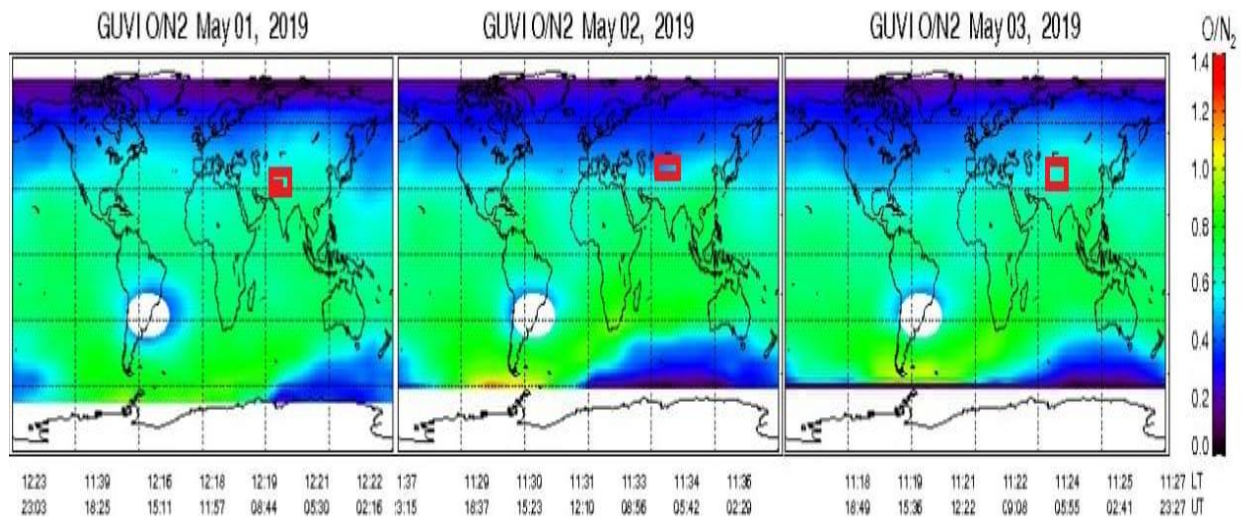


Figure 3.10: O/N₂ VALUES FROM MAY1, 2019 TO MAY 3, 2019

Electron density in the ionosphere is due to the presence of atomic oxygen, chemical reactions of N_2 and O^+ . Due to geomagnetic storms oxygen atoms present in the ionosphere are largely ionized and result in an increase in the O/N_2 ratio. Intensity of the storm was low on May 1, 2019 over the Sukkur region and suddenly increased on May 2, 2019 as Figure 3.10 shows O/N_2 data of three consecutive days MAY 1, 2019 to MAY 3, 2019. Sukkur region is highlighted in red color. We can see that on May 1, 2019 there are no abnormal clouds. A clear disturbance is shown on May 2, 2019 over the Sukkur region indicating the maximum strength of the storm. However, normal values were retrieved on May 3, 2019 by the atmosphere as we can see in figure 3.7. [48]

3.8 Day to Night data Comparison

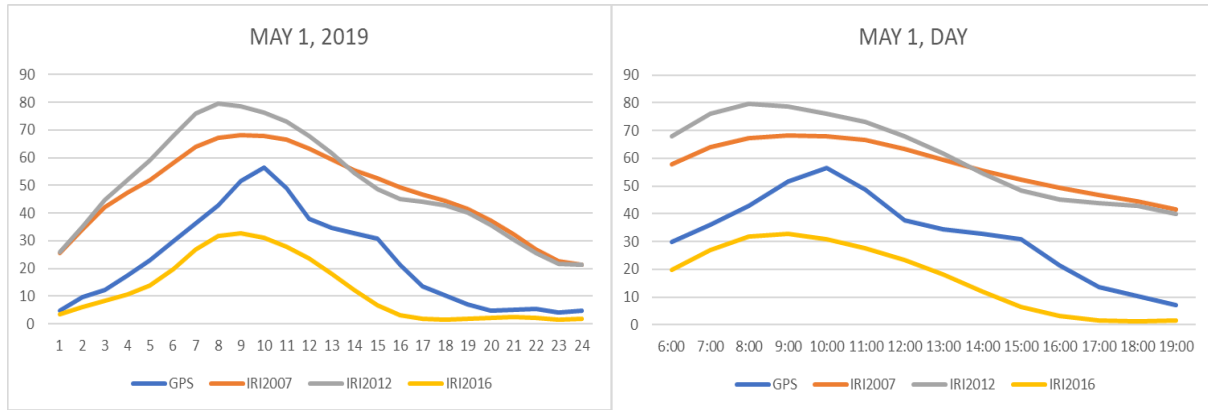


FIGURE 3.11a

FIGURE 3.11b

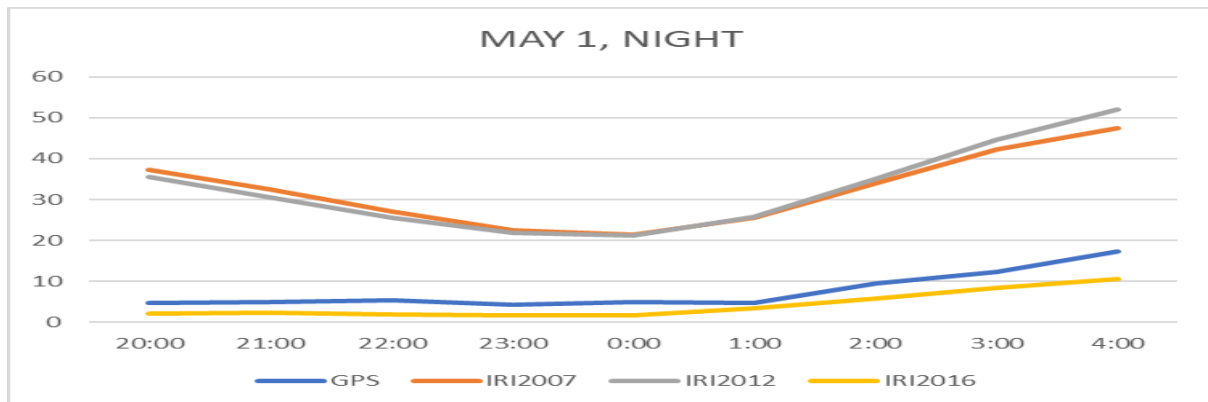


FIGURE 3.11c: TEC variations from GPS IRI-2007, 2012 and 2016 model over SUKKUR station on May 1, 2019.

Figure 3.11a is showing the TEC values of GPS and all IRI models on 1st May 2019. An increased in the TEC values is observed in sunrise and decreased in sunset time. Maximum value is observed at about 8am and minimum value is observed at about 8pm to 11pm approximately. Figure 3.11b is showing the data of daytime and figure 3.11c is showing nighttime data. Affirmation that sun has its impact on ionization.[49]

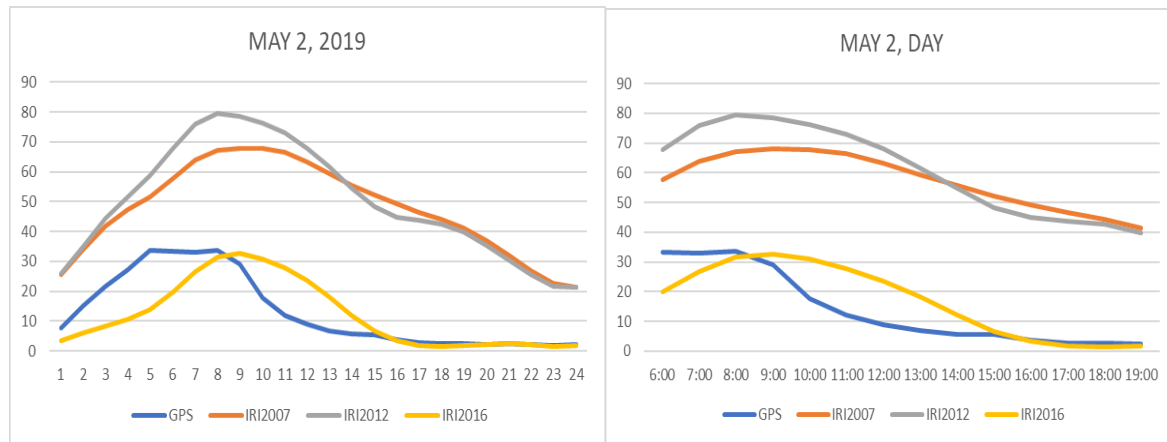


Figure: 3.12a

Figure: 3.12b

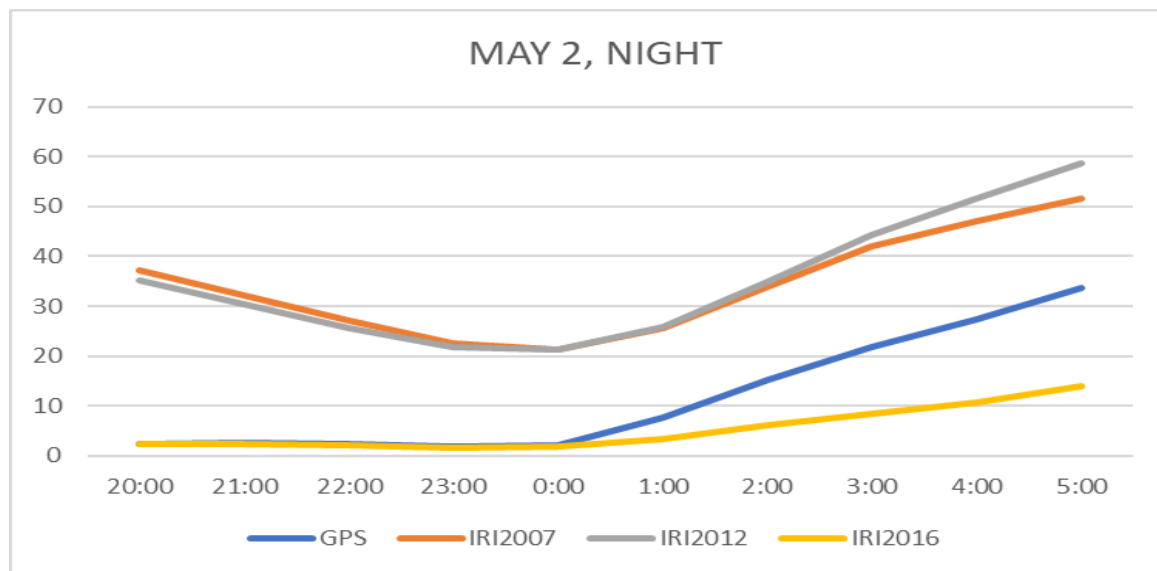


FIGURE 3.12c: TEC variations from GPS, IRI-2007,2012 and I2016 model over SUKKUR station on May 2, 2019.

Almost similar trend is observed in day 2 as Figure 3.12a is showing the TEC values of GPS and all IRI models on ^{2nd} May 2019. An increased in the TEC values is observed in sunrise and decreased in sunset time. Maximum value is observed at about 8am and minimum value is observed at about 8pm to 11pm approximately. Figure 3.12b is showing the data of daytime and figure 3.12c is showing nighttime data. [50]

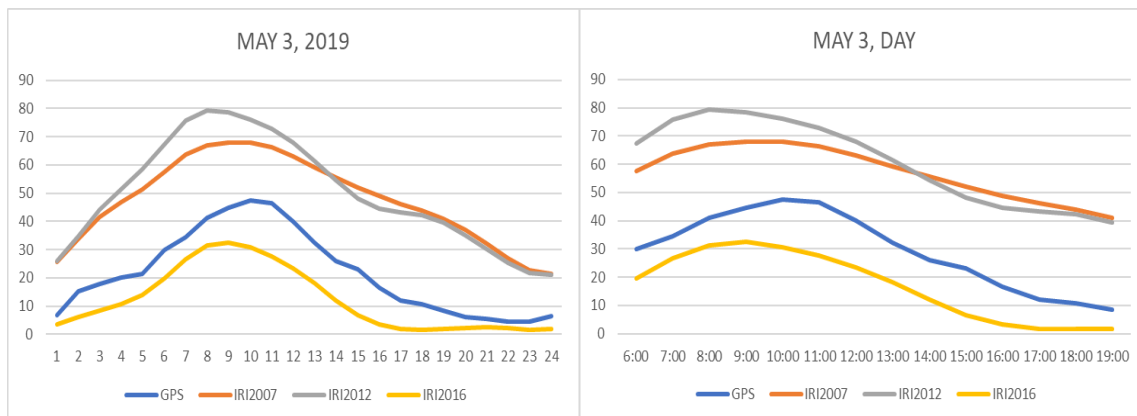


Figure: 3.13a

Figure: 3.13b

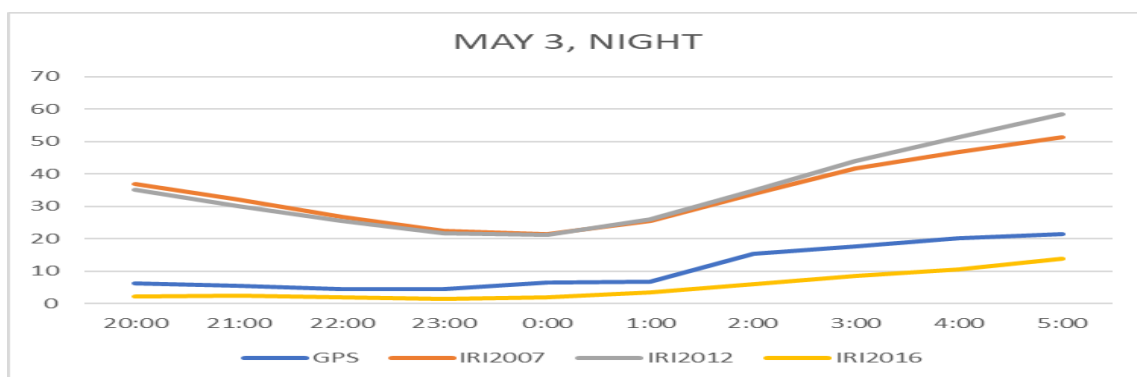


FIGURE 3.13c: TEC variations from GPS, IRI-2007, 2012 and 2016 model over SUKKUR station on May 3, 2019.

Almost similar kind of trend is observed in 3rd May 2019 TEC data of GPS and all other IRI models as shown in figure 3.9a, 3.9b and 3.9c.

4 Conclusions

I have compared GPS TEC with IRI 2007, IRI 2012 and IRI 2016 models over Sukkur region to check the correctness and reliability of the data gained from GPS and IRI models. Data of first three consecutive days of May 2019 was examined through different parameters. GPS TEC and IRI models are showing almost similar trend in all three days. A moderate storm is observed specially on May 2, 2019 as per different indices ($Dst = -30nT$ and $Kp = 40$). Comparison shows maximum value at sunrise of each day and minimum values at night as an affirmation that sun has its greater impact on atmospheric parameters. The results show that the IRI TEC values generally compared well with the GPS TEC values, with correlation coefficients as good as about 0.83 with IRI 2016, 0.82 with IRI 2012 and 0.78 with IRI 2007. Diurnal and hourly comparisons were generally better than the day to day comparisons since the IRI model is known to compare well with mean or median values. Moreover, it was observed that IRI 2016 is a good climate model that can provide almost accurate estimates of ionospheric delays of signals propagating over Pakistan. Light discrepancies are due to upper integration limit which was taken 1000km. As we know that upper integration limit is not constant, and the limits will not always correspond to 1000km as we have used in this work. Here is another possible reason for the discrepancies is in the relatively small data from the region considered in developing the model.

5 Future Work

- Data used in this work was for year 2019. I proposed doing similar work for the region and for different years specially in all years of high and moderate solar activity and when done, will enhance the practical understanding initiated in this work.
- A major interest in this work is to establish a rationale on how to use the IRI models to develop a TEC map for Pakistan. I suggest that similar work be done for other GPS stations across the country.

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